Discrete Math Class Notes

These are class notes for a first year undergrad Discrete Math class taught from Grimaldi's 'Discrete and Combinatorial Mathematics ($5^{th}$ edition)', which we will refer to as 'the text'.

# 1 Counting

This section draws from Sections 1.1 - 1.4 of the text. It is understood that the students have seen basic counting techniques, so we only review these topics quickly. Many of what would be examples are posed as problems.

**Topics Covered:**

- Rule of Sum
- Rule of Product
- Counting the Complement
- Permutations of sets and multisets
- Circular permutations
- Combinations
- Combinations with replacement

A *set* is a collection of objects, which we call *elements* or *members* of the set. For the time being, the sets we consider will be finite sets in which no member occurs more than once.

**Example 1.1.** The collection $S$ of two digit integers not divisible by 3 is a set:

$$S = \{10, 11, 13, 14, 16, 17, 19, \ldots, 97, 98\}$$

We are interested in counting $|S|$, the number of elements in $S$. Clearly we could just count, and find that there are 60. There are some basic techniques that allow us to simplify this task.

## 1.1 Basic Counting Techniques:

**The Rule of Sum (Divide and Conquer):** If we can break a set $A$ into smaller sets $A_1, \ldots, A_d$ such that every element is in exactly one set, then

$$|A| = |A_1| + \cdots + |A_d|.$$

For example, we could break the set $S$ from Example 1.1 up into the set $S_1 = \{10, 13, 16, 19, \ldots, 94, 97\}$ of the form $3x + 1$ for some integer $x$, and the set $S_2 = \{11, 14, 17, 20, \ldots, 95, 98\}$ of the form $3x + 2$. It is not hard to see that $|S_1| = |S_2|$ so we only have to count one of them to see that $|S| = |S_1| + |S_2| = 30 + 30 = 60$.

**The Rule of Product:** One good way of counting elements of a set is considering the process of choosing an element of the set. If the process can be broken down into independent steps, we can simply cound the ways of completing each step.

For example, the elements of $S$ are integers of the form $3x + y$ where $x$ is in $\{3, 4, \ldots, 32\}$ and $y$ is in $\{1, 2\}$. So we construct every element by choosing an $x$ and a $y$. We can choose $x$ in 30 ways and $y$ in 2 ways, so there are $30 \cdot 2 = 60$ ways of choosing an element of $S$.

**Counting the Complement:** Sometimes it is easy to count a bigger set and then remove items not in the wanted set.

For example, it is easier to count the set $T$ of integers between 1 and 99 that are not divisible by 3. There are 66 of them. The *complement* $\overline{S}$ of $S$ in $T$, that is the set of elements of $T$ not in $S$ is $\overline{S} = \{1, 2, 4, 5, 7, 8\}$, so $|S| = |T| - |\overline{S}| = 66 - 6 = 60$.

## 1.2 Permutations and Combinations

### 1.2.1 Permutations of a set

A *permutation* of a set is an arrangement of the items in a line. So $1, 2, 3$ and $3, 1, 2$ are different permutations of the set $\{1, 2, 3\}$. Using the rule of product, one can show that there are

$$n! = n \cdot (n - 1) \cdot \ldots \cdot 1$$

permutations of an $n$ element set. Indeed, there are $n$ ways to choose the first element, and then $n - 1$ ways to choose the second, etc. The notation '$n!$' is read '$n$ factorial'.

An $r$-*permutation* or a *size-$r$ permutation* of set is an arrangement of $r$ items of the set in a line. Again by the rule of product, one can show that there are

$$P(n,r) := \frac{n!}{(n-r)!}$$

$r$-permutations of an $n$ element set.

### 1.2.2 Permutations of a multiset

A *multiset* is a collection of elements that are not necessarily distinct. The multiset $\{1,2,2,2\}$ is different from $\{1,2\}$, though as sets, they would be the same. Counting permutations of a multiset adds a slight complication to the task of counting permutations of a set: certain permutations are indistinguishible, so we consider them the same. The multiset $\{1,2,2,2\}$ only has 4 distinguishible permutations:

$$\{(1,2,2,2),(2,1,2,2),(2,2,1,2),(2,2,2,1)\}.$$

This is less than the $4! = 24$ permutations of a 4-element set.

We could get this by counting the elements of the multiset as distinct, in $4!$ ways, and then observing that each distinguishible permutation of the multiset is counted $3!$ times, one for each permutation of the repeated elements. So there are $4 = 4!/3!$ distinguisible permutations of the multiset. More generally we have...

**Rule:** In a multiset containing $n_i$ *indistinguisible* elements of type $i$ for $i = 1,\ldots,r$, the number of *distinguishible* permutations is

$$\frac{(n_1+n_2+\cdots+n_r)!}{n_1!\cdot n_2!\cdot\cdots\cdot n_r!}.$$

When we talk of permutations of a multiset, it is understood that we are talking of distinguishible permutations, so we will generally drop the 'distinguishible.'

**Problem 1.1.** Find the number of permutations of the letters MISSISSIPPI.

**Problem 1.2.** Find the number of permutations of the letters ABCDEFGH in which the letters of the word CAGE occur in this order. In which the word CAGE doesn't occur.

**Problem 1.3.** Count the number of paths in the plane from $(0,0)$ to $(4,9)$, using only steps that go one unit up or 1 unit to the left

**Hint**

Text Example 1.14

A *circular permutation* of a set is an arrangement of the elements in a circle. Two such arrangements are considered the same if you can get from one to the other by rotating the circle. These are the permutations we are considering in the next problem.

**Problem 1.4.**

(a) How many ways can 6 people be seated around a circular table. (That is, how many circular permutations are there of a 6 element set?)

(b) How many ways can 3 men and 3 women be seated around a circular table so that men and women alternate.

(c) How many ways can 3 couples be seated around a table so that the couples are seated together.

## 1.3 Combinations without replacement

An $r$-combination of an $n$-element set is a choice of $r$ elements from the set. If the $r$-elements must be distinct, there are

$$\binom{n}{r} = P(n,r)/r! = \frac{n!}{(n-r)!r!}$$

of them, ( They are permutations in which order is disregarded.) The notation $\binom{n}{r}$ is read '$n$ choose $r$', and is called a *binomial co-efficient* because of the following theorem.

**Theorem 1.2** (Binomial Theorem)**.**

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^{n-i} y^i$$

*Proof.* Indeed, $(x+y)^n$ is the product of $n$ factors $(x+y)$. Every summand in the expansion of $(x+y)^n$ corresponds to a choice of either $x$ or $y$ from each of these terms. The summand $x^{n-i}y^i$ will occur once for every way we can choose $i$ of the $n$ terms to choose $y$ from. This is $\binom{n}{i}$ ways. □

Here is the typical problem.

**Example 1.3.** A pizza place has 16 different toppings. On a deluxe pizza you can choose 6 different toppings. How many different deluxe pizzas are there?

Solution

This is pretty straight forward: $\binom{16}{6}$.

See if you can deal with the following complications of this problem.

**Problem 1.5.** A pizza place offers the following toppings: 4 kinds of cheese, 7 kinds of vegetables, and 5 kinds of meat. How many different pizzas have

i. 2 cheeses, 4 vegetables and 3 meats (all different).

ii. 6 different toppings at least 3 of which are meat.

iii. 6 different toppings at least 3 of which are cheese.

Some problems can be dealt with as permutations or combination problems.

**Example 1.4.** How many ways can we split 20 people into 4 teams $A, B, C, D$ of 5 people each. (Teams are distinguishable.)

Solution

We can view this as a permutation problem or a combination problem. We can choose 5 for the first team, 5 for the second, etc., so there are

$$\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}.$$

Alternately, we can arrange 5 of each of $A, B, C, D$ among 20 people:

$$\frac{20!}{5!5!5!5!}$$

You can show that these quantities are the same.

**Example 1.5.** How many ways can you arrange the letters MISSISSIPPI such that there are no adjacent I's.

Solution

We can do this using combinations and permutations together. First arrange the letters MSSSSPP in $7!/(4!2!)$ ways. Given such an arrangement: SMSPSSP, there are 8 spaces in which we can put one I: -S-M-S-P-S-S-P-. There are $\binom{8}{3}$ ways to place the 3 I's in these spaces, so the total number of arrangements is

$$\frac{7!}{4!2!}\binom{8}{3}.$$

**Problem 1.6.** What is the co-efficient of $x^3y^3$ in

i. $(x+y)^6$ ?

ii. $(2x+y)^6$ ?

## 1.4 Combinations with replacement

Consider the following question:

**Example 1.6.** A doughnut store has kruellers, fritters and honey glazed doughnuts. How many ways can you select a dozen?

Observe that there is a one-to-one correspondence between the ways to select a dozen, and the arrangements of 12 stars: '$*$' and 2 dividers: '$|$'. The arrangement

$$** \,|\, * * * * * * * \,|\, * * *$$

partitions the 12 stars into three parts. This corresponds to the choice of 2 kruellers, 7 fritters, and 3 honey glazed. Thus there are $\binom{12+2}{2}$ choices of the dozen. It is clear that this also counts the solutions of

$$x_1 + x_2 + x_3 = 12$$

where $x_1, x_2, x_3$ are non-negative integers. The value of $x_i$ in a solution corresponds to the number of '$*$'s in the $i^{th}$ part.

Here are a couple of variations on the question.

**Problem 1.7.** How many solutions are there to the equation $x_1 + x_2 + x_3 = 12$ where all $x_i$ are non-negative integers and $x_2 \geq 2$?

**Hint**

(1.7) One can do this by counting the non-negative integer solutions of $x_1 + x_2' + x_3 = 10$ (where $x_2' = x_2 - 2$).

**Problem 1.8.** How many solutions are there to the equation $x_1 + x_2 + x_3 \leq 12$ where all $x_i$ are non-negative integers.

**Problem 1.9.** A composition of an integer $n$ is an ordered sum of positive integers totalling to $n$. ( So 8, $7+1$, $1+7$ and $1+1+1+1+1+1+1+1$ are all different compositions of 8. ) How many compositions are there of 8?

**Hint**

(1.8) One can do this by counting the non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 12$, (where $x_4$ is called a slack variable.)

**Problem 1.10.** How many ways can seven 'H's and nine 'T's be arranged in a line so that there exists exactly four strings of consecutive 'T's?

## 2 Symbolic Logic

This draws from Chapter 2 of the text.

Recall that an integer is prime if it is greater than one, and its only positive factors are one and itself. The following is a (sketch of a) well known proof of the fact that there are infinitely many primes.

*Proof.* Assume that there are only finitely many primes. Then we may list them: $p_1, p_2, \ldots, p_n$. Let $P = (p_1 \cdot p_2 \cdot \cdots \cdot p_n) + 1$. $P$ is greater than all of the $p_i$ so it is not prime. Since it is not prime, it can be written as the product of primes, (this fact is called the fundamental theorem of arithmetic). In particular, it can be written as $P = Qp_i$ for some $p_i$. But then $p_i$ divides $P$ and $P - 1$ so divides $P - (P - 1) = 1$. This is impossible, because $p_i \geq 2$. Our assumption that there are only finitely many prime must, therefore, be false. So there are infinitely many primes. $\square$

We wanted to prove a statement 'S is true.' but we instead assumed 'S is false' and showed this was inconsistent. This is called a 'proof by contradiction', and is a common proof technique. Similarily to prove an implication 'if $a$ then $b$', a useful approach is to show that 'if $b$ is not true, then $a$ is not true'. Why is this okay?

Symbolic logic is useful in deciding if two approaches to a proof are equivalent. We look at symbolic logic with a view to proof, techniques, but take it further than this.

**Topics Covered:**

- Truth value of primitive and compound statement through truth tables
- Tautology, contradiction, logical equivalence, laws of logic, rules of substitution
- Contrapositive, inverse, converse
- Valid argument, premise, conclusion, logical implication, rules of inference
- Quantification, open statement, universe of discourse

## 2.1 Basic Connectives and Truth Tables

**Definition 2.1.** A *statement* or *proposition* is a declarative sentence that is either true or false, not both, and it must be decidable which it is.

Examples of statements are '$1 + 1 = 3$', '7 is a quadratic residue modulo 10'. The sentence '$x > 0$' is not a statement, it is neither true or false, but if $x$ is defined as a number, then it becomes a statement.

Statements are usually denoted by lowercase letters. We might let $p$ be the statement $1+1 = 3$, in which case we could determine that $p$ is a false statement.

**Definition 2.2.** Given a statement $p$, the *negation* of $p$, or 'not $p$', is denoted $\neg p$.

Where $p$ is $1+1 = 3$, $\neg p$ is the statement '$1+1 \neq 3$', which is a true statement.

### 2.1.1 Basic Connectives

Negation is not the only way we can construct one statement from others. A statement that cannot be broken down into smaller statements is called a *primitive statement* or just a *primitive*.

Given primitives $p$ and $q$ we can construct *compound statements* using any of the following logical connectives. 'And' denoted $\wedge$; 'or' denoted $\vee$; 'xor' or 'exclusive or' denoted $\underline{\vee}$; 'implies' denoted $\rightarrow$; and 'biconditional' or 'if and only if' or 'iff' denoted $\leftrightarrow$.

Statements can be build up from other statements. For example, where $p$ is $1 + 1 = 3$ and $q$ is $1 + 1 = 2$, $p \vee q$ is the statement that '$1 + 1$ is 3 or 2', and is true because at least one of $p$ and $q$ was true.

The truth value, that is, the truth or falsity, of a compound statement depends only on the truth value of the primitives in it. The above connectives are thus rigourously defined by the following table, where we denote truth by 1 and falsity by 0.

| $p$ | $q$ | $\neg p$ | $p \wedge q$ | $p \vee q$ | $p \underline{\vee} q$ | $p \rightarrow q$ | $p \iff q$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |

This table is called a *truth table*. They can be used to determine the value of any compound statement from the value of their component primitives. For more complicated statements, we build up the truth table in several steps. The general approach is to approach a statement beginning with 'not' and then bracketed statements, and then working left to right.

**Problem 2.1.** Make a truth table for the statement $(\neg p \vee q) \to r$.

Solution

| $p$ | $q$ | $r$ | $\neg p$ | $\neg p \vee q$ | $\neg p \vee q \to r$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 1 |

Observe that in the above example, the statement was true in seven out of eight of the possible truth assignments of the primitives. One thing we are interested in is when a statement is always true, or always false, whatever the truth values of the primitives. From now on, we treat primitives as variable, which can be true or false. Though they are true or false, we don't know what they are.

**Definition 2.3.** A *tautalogy* is a statement that is always true. A *contradiction* is a statement that is always false.

**Example 2.4.** The statement $p \vee \neg p$ is a tautology and the statement $p \wedge \neg p$ is a contradiction.

**Problem 2.2.** Using a truth table, decide whether or not the statement $(p \to (q \to p))$ is a tautology or a contradicion. Can you argue this without a truth table?

One sees that if a statement has $n$ primitives, then the truth table will have $2^n$ rows. Three primitives is about the most we want to deal with. From here the theory of logical systems that we start to develop will allow us to decide if a statement is a truth or contradiction through more efficient means.

**Remark 2.5.** Denoting truth by 1 and falsity by 0 exemplifies the analogy between the logic algebra that we are developing, and a boolean algebra. We could further view $\veebar$ and $\wedge$ as analogous to addition and multiplication. We won't follow this analogue any more, but if you are familiar with boolean algebras, it might be a useful analogue.

## 2.2 Logical Equivalence

**Problem 2.3.** Construct truth tables for the statements '$p \to q$' and '$\neg p \vee q$'.

For every truth assignment of the primitives $p$ and $q$, these two compound statements have the same truth value.

**Definition 2.6.** Statements $s_1$ and $s_2$ are *logically equivalent* if $s_1$ is true if and only if $s_2$ is true. We denote this $s_1 \iff s_2$. If $p$ is a tautalogy, we write $p \iff T_0$ and if it is a contradiction, we write $p \iff F_0$.

**Remark 2.7.** The notations '$s_1 \leftrightarrow s_2$' and '$s_1 \iff s_2$' mean different things. The first is a logical statement- it may be true or false, and writing it down we are making no assertion about its truth value. The second, '$s_1 \iff s_2$', says that the first, '$s_1 \leftrightarrow s_2$', is a tautalogy.

It is easy to show (though not always easy to determine) that non-equivalent statements are non-equivalent. One only has to provide one line of the truth table- a true assignment of the primitives in which the statements differ.

**Example 2.8.** Show that $p \wedge (\neg p \to q)$ is not a tautalogy.

Solution

Where $p = 1$ and $q = 0$ we have $\neg p = 1$ so $(\neg p \to q) = 0$ so $p \wedge (\neg p \to q) = 1 \wedge 0 = 0$. Thus the statement is not a tautalogy.

To prove two statements are equivalent, we can always use a truth table, but we would like to avoid doing this too often. So we build up a list of equivalences that we can use over and over.

Above, we saw that

$$p \to q \iff \neg p \vee q$$

so '$\to$' can be expressed in terms of '$\neg$' and '$\vee$'. In the same way, we have

$$\begin{aligned} p \leftrightarrow q \quad &\iff \quad (p \to q) \wedge (q \to p) \\ &\iff \quad (\neg p \vee q) \wedge (\neg q \vee p) \\ &\iff \quad (\neg p \wedge \neg q) \vee (p \wedge q) \end{aligned}$$

and

$$p \underline{\vee} q \iff (p \vee q) \wedge \neg(p \wedge q).$$

Often we will use these equivalences to reduce compound statements to statements using only $\neg$, $\vee$ and $\wedge$.

**Problem 2.4.** Convince yourself of the logical equivalences given above. Prove that $\wedge$ can also be expressed in terms of '$\neg$' and '$\vee$'.

There are many other useful logical equivalences

**Problem 2.5.** Use a truth table to show that

$$\neg(p \wedge q) \iff (\neg p \vee \neg q).$$

This is called DeMorgan's law.

We will frequently use the following Laws of Logic.

| | | |
|---|---|---|
| 1 | $\neg\neg p \iff p$ | Double Negation |
| 2 | $\neg(p \vee q) \iff \neg p \wedge \neg q$<br>$\neg(p \wedge q) \iff \neg p \vee \neg q$ | Demorgan |
| 3 | $p \vee q \iff q \vee p$<br>$p \wedge q \iff q \wedge p$ | Commutativity |
| 4 | $p \vee (q \vee r) \iff (p \vee q) \vee r$<br>$p \wedge (q \wedge r) \iff (p \wedge q) \wedge r$ | Associativity |
| 5 | $p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$<br>$p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$ | Distributivity |
| 6 | $p \vee p \iff p$<br>$p \wedge p \iff p$ | Idempotence<br>Idempotence |
| 7 | $p \vee F_0 \iff p$<br>$p \wedge T_0 \iff p$ | Identity |
| 8 | $p \vee \neg p \iff T_0$<br>$p \wedge \neg p \iff F_0$ | Inverse |
| 9 | $p \vee T_0 \iff T_0$<br>$p \wedge F_0 \iff F_0$ | Domination |
| 10 | $p \vee (p \wedge q) \iff p$<br>$p \wedge (p \vee q) \iff p$ | Absorption |
| (11) | $p \to q \iff \neg p \vee q$ | Implication |

**Problem 2.6.** Read through each of the laws of logic, and convince yourself that they are true. Prove your favourite one using a truth table. Prove that it is your favourite one. Prove that it doesn't pay to choose favourites.

### 2.2.1 Substitution Rules

On top of the laws of logic, we have some substitution rules which allow us to use them more easily.

- If a statement $s$ is a tautalogy, and every occurence of some primitive $p$ in $s$ is replaced by another statement $t$, then the resulting statement is a tautology.
- If $s \iff t$, you can always replace $s$ with $t$ in statement without changing the truth values.

**Example 2.9.** The statement $s : (p \to q) \leftrightarrow (\neg p \vee q)$ is a tautology. Replacing $p$ with $(\neg r \wedge t)$ yields that

$$((\neg r \wedge t) \to q) \leftrightarrow (\neg(\neg r \wedge t) \vee q)$$

is also a tautology.

**Example 2.10.** By the law of double negation we have:

$$(p \to q) \to \neg p \iff (\neg\neg p \to q) \to \neg p.$$

Here is a more meaty example of the use of the Laws of Logic and the Rules of substitution.

**Example 2.11.** Show that $(\neg p \to q) \to r \iff (p \to r) \wedge (q \to r)$.

Solution

| | | |
|---|---|---|
| | $(\neg p \to q) \to r$ | |
| $\iff$ | $(\neg\neg p \vee q) \to r$ | Substituting $\neg p$ for $s$ and $q$ for $t$ into $s \to t \iff \neg s \vee t$ |
| $\iff$ | $(p \vee q) \to r$ | Double Negation |
| $\iff$ | $\neg(p \vee q) \vee r$ | Subtitution into $s \to t \iff \neg s \vee t$ |
| $\iff$ | $(\neg p \wedge \neg q) \vee r$ | DeMorgan |
| $\iff$ | $r \vee (\neg p \wedge \neg q)$ | Commutativity |
| $\iff$ | $(r \vee \neg p) \wedge (r \vee \neg q)$ | Distributivity |
| $\iff$ | $(p \to r) \wedge (q \to r)$ | $s \to t \iff \neg s \vee t$ |

There are a couple more statements that are related to the implication $p \to q$ which get special names as they are used often as proof techniques.

The *contrapositive* of $p \to q$ is $\neg q \to \neg p$. The *converse* of $p \to q$ is $q \to p$. The *inverse* of $p \to q$ is $\neg p \to \neg q$.

**Problem 2.7.** Show that an implication is logically equivalent to its contrapositive, but not to its converse or inverse. What does this tell you of the relation between the converse and the inverse of an implication?

In proving an implication, one common technique is to prove the contrapositive. When we prove an implication, we often comment on whether or not the converse is true.

**Problem 2.8.** Where $p$ is the sentence 'It is raining', and $q$ is the sentence 'I skip class', what are the English interpretations of $p \to q$, its contrapositive, its inverse, and its converse.

Let's finish this section with another example of proving the equivalence of statements using our laws of logic.

**Example 2.12.** Negate and simplify the statement $p \to (\neg q \wedge r)$.

Solution

Leaving the word 'simplify' a little vague, we proceed:

$$\begin{array}{lcl} & \neg(p \to (\neg q \wedge r) & \\ \iff & \neg(\neg p \vee (\neg q \wedge r) & \text{Using } s \to t \iff \neg s \vee t \\ \iff & \neg\neg p \wedge \neg(\neg q \wedge r) & \text{DeMorgan} \\ \iff & p \wedge (\neg\neg q \vee \neg r) & \text{Double Negation then DeMorgan} \\ \iff & p \wedge (r \to q) & \text{Using } s \to t \iff \neg s \vee t \end{array}$$

Observe that we used Double Negation and DeMorgan together in one step, and then in the next step, we used Double Negation without even mentioning it. We allow such shortcuts if it doesn't hinder reading. But don't take them too far.

The text finished this section with some examples of how a logical statement can be modelled by an electronic circuit. We won't look at it, but it you are so inclined, it is an interesting way to visualise the equivalence of statements.

## 2.3 Inference Rules

### 2.3.1 Logical Implication

**Definition 2.13.** An *argument* consists of a conjunction of statements :

$$p_1 \wedge p_2 \wedge \cdots \wedge p_d$$

called the *premises* and a *conclusion* $q$.

We write an argument as

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_d] \to q.$$

It is called true or *valid* if it is a tautalogy.

**Example 2.14.** Is the following argument valid?

$$[(p \to r) \wedge (\neg q \to p) \wedge \neg r] \to q$$

Solution

Again, we can use a truth table.

| $p$ | $q$ | $r$ | $s_1 : p \to r$ | $s_2 : \neg q \to r$ | $s_3 : \neg r$ | $[s_1 \wedge s_2 \wedge s_3] \to q$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | |
| 0 | 0 | 1 | 1 | 1 | 0 | |
| 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | |
| 1 | 0 | 0 | 0 | 0 | 1 | |
| 1 | 0 | 1 | 1 | 1 | 0 | |
| 1 | 1 | 0 | 0 | 1 | 1 | |
| 1 | 1 | 1 | 1 | 1 | 0 | |

There is only one row in which all of the premises are true, and in this case the conclusion is also true, so the argument is valid.

**Example 2.15.** Is the argument $[(p \to q) \wedge q] \to p$ valid?

Solution

No! (Don't you dare.) If $q = 1$ and $p = 0$ then $(p \to q) \wedge q$ is 1, but $p = 0$.

**Example 2.16.** Is the argument $[p \vee \neg p] \to q$ valid?

Indeed it is, as the premises are a contradiction. In this case we say that the argument is *vacuoulsy true*. 'Vacuous' means 'empty', like a vacuum. It is saying, 'sure the argument is true, but this is not really saying anything'.

**Definition 2.17.** If $p$ and $q$ are statements such that $p \to q$ is a tautology, then we say $p$ *logically implies* $q$ and write $p \Rightarrow q$

Note that $p \to q$ can be true or false, but $p \Rightarrow q$ means $p \to q$ is a tautology. So '$p \iff q$' is the same as '$p \Rightarrow q$ and $q \Rightarrow p$'.

**Problem 2.9.** Use a truth table to prove that $[(p \to q) \wedge p] \Rightarrow q$.

The logical implication in the above problem is a *inference rule* called *Modus Ponens*. Having proved it, we use it in proving the validity of other arguments.

**Example 2.18.** To prove the implication $[(p \to q) \wedge \neg q] \Rightarrow \neg p$ we start with the premises, and work to the conclusion.

$$\begin{array}{rcl} & (p \to q) \wedge \neg q & \\ \iff & (\neg q \to \neg p) \wedge \neg q & (contraposative) \\ \Rightarrow & \neg p & (ModusPonens). \end{array}$$

It will make an argument clearer if we write it in a table as follows.

| | | |
|---|---|---|
| 1 | $p \to q$ | premise (given) |
| 2 | $\neg q$ | premise |
| 3 | $\neg q \to \neg p$ | contraposative of 1 |
| $\therefore$ | $\neg p$ | 2 and 3 and Modus Ponens |

The symbol $\therefore$ is read 'therefore', and denotes that we have arrived at a conclusion. The implication above is another inference fule called *Modus Tollens*. We have several more inference rules:

| | Premises | Conclusion | Name |
|---|---|---|---|
| 1 | $p \wedge (p \to q)$ | $q$ | Modus Ponens |
| 2 | $(p \to q) \wedge (q \to r)$ | $p \to r$ | Syllogism |
| 3 | $(p \to q) \wedge \neg q$ | $\neg p$ | Modus Tollens |
| 4 | $(p) \wedge (q)$ | $p \wedge q$ | Conjunction |
| 5 | $(p \vee q) \wedge \neg p$ | $q$ | Disjunctive Syllogism |
| 6 | $(\neg p \to F_0)$ | $p$ | Contradiction |
| 7 | $p \wedge q$ | $p$ | Conjunctive Simplification |
| 8 | $p$ | $p \vee q$ | Disjunctive Amplification |
| 9 | $(p \wedge q) \wedge [p \to (q \to r)]$ | $r$ | Conditional Proof |
| 10 | $(p \to r) \wedge (q \to r)$ | $(p \vee q) \to r$ | Proof by cases |
| 11 | $(p \to q) \wedge (r \to s) \wedge (p \vee r)$ | $q \vee s$ | Constructive Dilemma |
| 12 | $(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)$ | $(\neg p \vee \neg r)$ | Destructive Dilemma |

These will help us prove logical implications.

**Example 2.19.** Show the argument $[(p \to q) \wedge \neg q \wedge \neg r] \to \neg(p \vee r)$ is valid.

Solution

| | | |
|---|---|---|
| 1 | $p \to q$ | premise |
| 2 | $\neg q$ | premise |
| 4 | $\neg p$ | 1 and 2 and Modus Tollens |
| 5 | $\neg r$ | premise |
| 6 | $\neg p \wedge \neg r$ | conjunction of 4 and 5 |
| $\therefore$ | $\neg(p \vee r)$ | 6 DeMorgan |

**Problem 2.10.** Prove $[(p \to q) \wedge (r \to \neg q) \wedge r] \Rightarrow \neg p$.

**Problem 2.11.** Prove $[(p \to (q \to r)) \wedge (p \vee s) \wedge (t \to q) \wedge \neg s] \Rightarrow (\neg r \to \neg t)$.

To show that an argument is false, we need only to give a counterexample.

**Example 2.20.** To see that

| | |
|---|---|
| 1 | $p \to q$ |
| 2 | $\neg p$ |
| $\therefore$ | $\neg q$ |

is an invalid argument, let $p = 0$ and $q = 1$. Then premises 1 and 2 are true, but the conclusion is false.

### 2.3.2 Common Proof Techniques

Two common proof techniques that we get from our inference rules are 'Proof by Contradiction' and 'Indirect proof.' Proof by Contradiction us a proof that used the rule of contradiction. For this we introduce a false premise, the negation of our conclusion, and show it leads to falsity.

**Example 2.21.** We show that

| | |
|---|---|
| 1 | $(s \wedge h) \to r$ |
| 2 | $r \to c$ |
| 3 | $\neg c$ |
| $\therefore$ | $\neg s \wedge \neg h$ |

is valid. Indeed, taking 1 - 3 as premises, we continues as follows

| | | |
|---|---|---|
| 4 | $\neg(\neg s \wedge \neg h)$ | contradictory premise |
| 5 | $s \wedge h$ | 4 and DeMorgan |
| 6 | $r$ | 1 and 5 and MP (Modus Ponens) |
| 7 | $c$ | 2 and 6 and MP |
| 8 | $c \wedge \neg c$ | 3 conjunction 7 |
| 9 | $F_0$ | 8 and Inverse Law |
| $\therefore$ | $\neg s \wedge \neg h$ | 4 and 9 and contradiction |

Since $(p \to (q \to r)) \iff ((p \wedge q) \to r)$, to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n] \to (q \to r)$$

it is enough to show

$$[p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge q] \to (r).$$

This is called an indirect proof.

**Example 2.22.** We prove the Rule of Syllogism (see the table) by indirect proof.

| | | |
|---|---|---|
| 1 | $p \to q$ | premise |
| 2 | $q \to r$ | premise |
| 3 | $p$ | indirect premise |
| 4 | $q$ | 1,3, MP |
| $\therefore$ | $r$ | 2,4 MP |

Thus by indirect proof, we get the rule of Syllogism.

**Problem 2.12.** Prove

| | |
|---|---|
| 1 | $p \to (q \to r)$ |
| 2 | $\neg q \to \neg p$ |
| $\therefore$ | $p \to r$ |

## 2.4 Quantifiers

Recall that statements such as

$$p(x) : x \text{ is a multiple of } 3,$$

and

$$q(x, y) : x > y,$$

are not **statements**. Because of the variable, their truth values cannot be determined. However, when their variables are replaced with numbers, the become statements.

**Definition 2.23.** An *open statement* is a sentence that

- contains one or more variable,
- becomes a statement when the variables are replaced with certain allowable choices.

**Example 2.24.** Where $p(x)$ is the open statement above, $p(6)$ is the true statement '6 is a mutliple of 3', while $p(5)$ is the false statement '5 is a mutliple of 3'.

There is another way to change a open statement into a statement.

**Definition 2.25.** The allowable choices for the variables of an open statement are called the *universe* or *universe of discourse*, of the open statement. The symbol $\forall$, read as 'for every', is the *universal quantifier*. The symbol $\exists$, read as 'for some', is the *existential quantifiers*. They *quantify* an open statement $r(x)$ with universe $U$, as follows.

'$\forall x[r(x)]$' is the statement that '$r(x)$ is true for **all** $x \in U$.'

'$\exists x[r(x)]$' is the statement that '$r(x)$ is true for **some** $x \in U$.'

Note

Often, the universe of an open statement does not matter, or can be assumed from 'what is reasonable'. Certainly to interpret $p(6)$ as the statement '6 is a multiple of 3', we don't need to know the universe. But for quantified statements, it is usually important. Indeed, where $p(x)$ is as above $\forall x[p(x)]$ is false if the universe is $U = \mathbb{Z}$, but true if it is $U = 3\mathbb{Z}$.

That said, choosing the universe as $U = 3\mathbb{Z}$ seems a little like trickery. Sometimes even for open statements, if the universe is omitted, we assume it to be the largest reasonable universe. The statement '$x$ is a multiple of 3' appears very much to be about the integers, it would be reasonable to assume this is the universe.

**Example 2.26.** Let $q(x, y)$ be the statement $x > y$ have universe $\mathbb{R}$. Then

(a) $\forall x \forall y[q(x, y)]$ is false.

(b) $\forall x \exists y[q(x, y)]$ is true.

(c) $\exists x \forall y[q(x, y)]$ is false.

Hey look! The order of the quantifiers matters.

**Problem 2.13.** Where the universe is $\mathbb{Z}$, let

- $p(x) : x^2 - 5x + 6 = 0$
- $q(x) : x > 0$
- $r(x) : x^2 - 7 \geq 0$.

Determine the truth value of the following statements.

(a) $\exists x[p(x) \wedge \neg q(x)]$

(b) $\exists x[\neg q(x) \rightarrow p(x)]$

(c) $\forall x[p(x) \rightarrow q(x)]$

(d) $\forall x[p(x) \vee (q(x) \wedge r(x))]$

**Problem 2.14.** Where the universe of $x$ is the set of students in the class, and the universe of $y$ is a score, let $s(x, y)$ be the open statement:

> $x$ got $y$ on the test.

**Hint**

You can use arithmetic symbols $<$, $\leq$, and $=$.

Express the following statements symbolically.

(a) Some student got 7.5 on the test.

(b) Every student got at least 5 on the test.

(c) Some student got more than 9 on the test.

(d) No student got 8.5 on the test.

(e) Only one student got 0 on the test.

### 2.4.1 Quantifications of Logical Equivalences

Recall the logical equivalence

$$p \to q \iff \neg p \vee q.$$

It follows by substitution that if $p(x)$ and $q(x)$ are open statements with the same universe, then

$$p(x) \to q(x) \iff \neg p(x) \vee q(x)$$

for any choice of $x$ in the universe.

Thus the following quantified versions of the logical equivalence are also true:

- $\forall x[p(x) \to q(x)] \iff \forall x[\neg p(x) \vee q(x)]$
- $\exists x[p(x) \to q(x)] \iff \exists x[\neg p(x) \vee q(x)]$

Every logical equivalence has similarily quantified versions.

### 2.4.2 Interaction of Quantifiers and Connectives

One can show the following.

**Proposition 2.27.** *(a)* $\forall x[p(x) \wedge q(x)] \iff \forall x[p(x)] \wedge \forall x[q(x)]$

*(b)* $\exists x[p(x) \wedge q(x)] \Rightarrow \exists x[p(x)] \wedge \exists x[q(x)]$

*(c)* $\forall x[p(x) \vee q(x)] \Leftarrow \forall x[p(x)] \vee \forall x[q(x)]$

*(d)* $\exists x[p(x) \vee q(x)] \iff \exists x[p(x)] \vee \exists x[q(x)]$

*(e)* $\neg\forall x[p(x)] \iff \exists x[\neg p(x)]$

*(f)* $\neg\exists x[p(x)] \iff \forall x[\neg p(x)]$

**Problem 2.15.** Convince yourself that the logical implications from Proposition 2.27 are true. Show that $\Leftarrow$ is not true in (b) and $\Rightarrow$ is not true in (c).

The above rules allow us to prove other logical implications involving the quantifiers.

**Example 2.28.** Show that $\exists x[p(x) \to q(x)] \wedge \forall x[p(x)] \Rightarrow \exists x[q(x)]$.

Solution

$$
\begin{array}{lcll}
 & & \exists x[p(x) \to q(x)] \land \forall x[p(x)] & \\
\iff & & \exists x[\neg p(x) \lor q(x)] \land \forall x[p(x)] & \text{from } s \to t \iff \neg s \lor t \\
\iff & & \exists x[\neg p(x)] \lor \exists x[q(x)] \land \forall x[p(x)] & \text{Prop 2.27 (d)}
\end{array}
$$

# 3 Set Theory

We cover Section 3.1-3.4 of the text, however have rearranged material so our first three sections don't agree exactly with the text. We also include material from Section 5.2 and 5.5 of the text in our Section 3.3.

**Topics Covered:**

- Set-builder notation, subset, emptyset
- Set operations, set laws, Venn diagrams
- Cardinality, power-set, inclusion-exclusion
- The Pigeonhole principle, counting by bijections
- Probability

## 3.1 Sets and Subsets

This mostly agrees with Section 3.1 of the text.

### 3.1.1 Basic Definitions

The notation of symbolic logic allows us to give precise definitions for many notions we have until now defined only intuitively.

We still leave the idea of a set $A$ a bit imprecise: 'a collection of distinct elements from a universe', but given this, we lets look at some related definitions.

Given two sets $A$ and $B$ with the same universe $U$, $A$ *is a subset of* $B$, written $A \subseteq B$, if

$$\forall x[x \in A \to x \in B]$$

.

**Problem 3.1.** Given this definition, is $A \subseteq A$. Where $\emptyset = \{\,\}$ is *the empty set*, is $\emptyset \subseteq A$? Is $\{3, 1, 2, 2\} \subseteq \{1, 2, 3\}$.

How would you define the statement $A = B$, that the sets $A$ and $B$ are the same set? One way would be

$$\forall x[x \in A \iff x \in B]$$

.

But look at the following logical equivalence:

$$\begin{aligned}\forall x[x \in A \iff x \in B] \quad &\iff \quad \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in A] \\ &\iff \quad A \subseteq B \wedge B \subseteq A\end{aligned}$$

This rigourously proves the obvious statement that

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

As simple as this seems, we will use it a lot in proofs.

A set $A$ is a *proper subset* of $B$, written $A \subset B$ if $A \subseteq B$ and $A \neq B$.

Give and alternate definition of $A \subset B$:

$A \subseteq B$ and _____.

As the basic set definitions have 'quantified open statement' versions, proofs about set theory will be very similar to proofs of logical equivalences and logical implications. Let's see one.

**Theorem 3.1.** *Let $A, B, C \subseteq U$.*

*i. If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.*

*ii. If $A \subset B$ and $B \subseteq C$ then $A \subset C$.*

*iii. If $A \subseteq B$ and $B \subset C$ then $A \subset C$.*

*iv. If $A \subset B$ and $B \subset C$ then $A \subset C$.*

*Proof.* We prove only *i* and *iii*. To prove *i*, assume the premises, that $A \subseteq B$ and $B \subseteq C$. By definition this means that

$$\begin{aligned} & \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in C] \\ \iff \quad & \forall x[(x \in A \to x \in B) \wedge (x \in B \to x \in C)] \\ \Rightarrow \quad & \forall x[(x \in A \to x \in C] \text{ by syllogism.}\end{aligned}$$

This last statement is the definition of $A \subseteq B$.

Now we prove *iii*. Assume that $A \subseteq B$ and $B \subset C$. As $B \subset C$ implies $B \subseteq C$ we have the premises of part *iii*., so we may conclude that $A \subseteq C$. Thus we only have to show that there is some $x$ that is in $C$ but not $A$. By the premise $B \subset C$ there is some $x$ in $C$ that is not in $B$. As $A$ is a subset of $B$, this $x$ is not in $A$ either. So we are done. $\square$

Notice that our proof style differed from part *i* to part *c*. In *a* we were very notational while in *c* we relied a lot more on english sentences. Which one is easier for you to read? Mathematics tends to be written as in part *c*, as it is easier to read. But it is still precise, many words or phrases have more restricted, or different usage in Math than they do in regular conversation. It is good to learn some of these usages.

**Note on English**

The word 'assume' is usually comes at the beginning of a proof and will be followed, often much later, by

- a conclusion, if we are proving an argument to be valid; or
- a contradiction, when we are doing proof by contradiction.

When the proofs involved between the assumptions and the conclusion/contradiction are very short, we often use 'if' instead of 'assume'. Ex. 'If $x$ is in $A$ then it is also in $B$, which is false, so $x$ is not in $A$.'

**Problem 3.2.** Finish the proof of Theorem 3.1. Try to write it more like the proof of part *iii* above.

### 3.1.2 Set-builder Notation

As for these definitions, universal quantification is usefel in describing specific sets. This is seen With *set-builder* notation. Given a universe $U$ and an open statement $p(x)$,

$$\{x \in U \mid p(x)\}$$

(or simply $\{x \mid p(x)\}$ when the universe is understood), is the set of elements in $U$ for which $p$ is true.

For example the set $2\mathbb{N} = \{0, 2, 4, 6, \dots\}$ can be described by $2\mathbb{N} = \{x \in \mathbb{Z} \mid x \geq 0, 2 \text{ divides } x\}$.

As a variation that works when the universe is understood, one can even use a function of $x$: $2\mathbb{N} = \{2x \mid x \geq 0\}$.

## 3.2 Set Operations and Laws of Set Theory

Mostly Section 3.2 and 3.3 of the text.

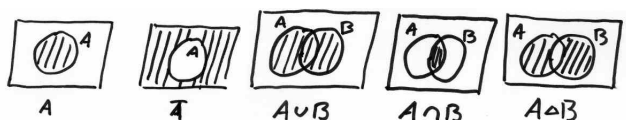
Each of the basic logical connectives defines an operation on sets.

Let $A$ and $B$ be sets with the same universe $U$, then we have

- the *complement* $\overline{A} = \{x \mid x \notin A\}$
- the *union* $A \cup B = \{x \mid x \in A \vee x \in B\}$

- the *intersection* $A \cap B = \{x \mid x \in A \wedge x \in B\}$, and
- the *symmetric difference* $A \triangle B = \{x \mid x \in A \underline{\vee} x \in B\}$.

These set operations can be visualised by a *Venn (or circle) Diagram.*




There is also the *relative complement of $A$ in $B$* defined as

$$A - B = \{x \mid x \in A \vee x \notin B\}.$$

**Problem 3.3.** Prove that $A - B = A \cap \overline{B}$

**Problem 3.4.** What is the set $\{x \mid x \in A \to x \in B\}$? Perhaps use the implication law. Draw a Venn diagram to describe it. How does this relate to $A - B$.

As the sets operations are defined from logical connectives, it makes sense that they exhibit similar properties, such as DeMorgan's Law:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

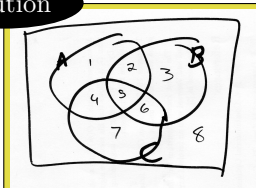
The proof of this follows directly from DeMorgans Law for logic:

$$\begin{aligned}
\overline{A \cup B} &= \overline{\{x \mid x \in A \vee x \in B\}} && \text{def of union} \\
&= \{x \mid \neg(x \in A \vee x \in B)\} && \text{def of complement} \\
&= \{x \mid x \notin A \wedge x \notin B\} && \text{DeMorgan} \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} && \text{def of complement} \\
&= \overline{A} \cap \overline{B}
\end{aligned}$$

Similarily, we get the Set Laws, set theoretic versions of all of the laws of logic. See the table on page 139 of the text. It is useful to also include the identity from Problem 3.3, which we view as an analogue of the implication law, in this list.

**Example 3.2.** Show that $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution




One way to do this is to use a Venn diagram. One can check that both sides of the equation are region 1 in the figure on the left. Alternately we can use the set laws.

$$\begin{aligned} A-(B\cup C) &= A\cap\overline{B\cup C} && \text{Implication Law}\\ &= A\cap\overline{B}\cap\overline{C} && \text{DeMorgan}\\ &= A\cap A\cap\overline{B}\cap\overline{C} && \text{Idempotence}\\ &= (A\cap\overline{B})\cap(A\cap\overline{C}) && \text{Commutativity}\\ &= (A-B)\cap(A-C) && \text{Implication}\end{aligned}$$

## 3.3 More on counting

This pulls from Sections 3.1, 5.2, 5.5

### 3.3.1 Double-Counting

The *cardinality* $|A|$ of a finite set $A$ is the number of elements in the set. For infinite sets such as $\mathbb{Z}$ or $\mathbb{R}$, we say the cardinality is $\infty$. In other classes you will see that there are various infinite cardinalities, but that won't interest us in this class.

The power set of $\mathscr{P}(A)$ of a set $A$ is the set of all subsets of $A$. If $|A| = n$, what is $|\mathscr{P}(A)|$?

One easy way to count it is using the product rule. In choosing an element of $\mathscr{P}(A)$, that is, a subset $S$ of $A$, we have to choose for each of the $n$ elements $x$ of $A$ whether or not $x$ is in $S$. There are $2^n$ ways to do this, so $\mathscr{P}(A) = 2^n$.

Another way to count is to use the sum rule. Counting the 0 element sets, then the 1 element sets, et cetera, we get $|\mathscr{P}(A)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$. Thus

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

The proof we have given of this identity is an example of what is called *double counting*. We counted a set in two different ways to arrive at the identity. This is a common technique for proving combinatorial indentities.

**Problem 3.5.** Show that this identity also follows from the Binomial theorem.

**Problem 3.6.** Show that

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$$

by double counting the number of $r$-combinations of an $n+1$ element set.

### 3.3.2 Counting by Bijections

We are good at counting permutations and combinations of various types. It makes sense to relate other counting problems to these problems. We saw this when counting the number of positive integer solutions to the equation $x_1 + x_2 + x_3 = 10$, but it is a fundamental idea.

Given two sets $A$ and $B$, if we can define a *bijection* ( a 1-to-1, onto function) from $A$ to $B$. Then $|A| = |B|$.

Even when we know the cardinality of a set $B$ it can be useful to get a bijection with another well know set $A$.

**Example 3.3.** We define a bijection $f$ from the powerset $\mathscr{P}([n]$ to the set of $n$ digit binary strings by letting the $i^{th}$ digit of $S \in \mathscr{P}([n])$ be 1 if $i \in S$ and 0 otherwise.

We can verify that it is a bijection by exhibiting an *inverse* function $g$ from the set of $n$-digit binary strings to $\mathscr{P}([n])$: $g(b_n b_{n-1} \ldots b_2 b_1)$ contains the element $i$ if and only if $b_i$ is 1.

Why is this useful, now we have an *ordering* of $\mathscr{P}([n])$ by converting these strings to the integers they represent. If we want to write an algorithm checking some property of subsets of $[n]$, this is useful.

A *composition* of the number $n$ is an expression of $n$ as a sum of positive integers. So $5 = 1+1+1+1+1$ , $5 = 1+2+1+1$, $5 = 2+1+1+1$, $5 = 4+1$. How many compositions are there of the number $n$?

We could count the number of compositons with $i$ summands for each $i$, and then add them up. (See Section 1.4 of the text.) But we will count the set by bijections.

Indeed, we will show that there are $2^{n-1}$ compositions of $n$ by giving a bijection between the set of compositions, and $\mathscr{P}([n-1]$.

Start with the string $(1+1+\cdots+1+1)$. There are $n-1$ occurences of the symbol '+'. For a set $S \in \mathscr{P}([n-1])$, let $f(S)$ be the composition constructed as follows. If $i$ is in $S$ replace the $i^{th}$ occurence of '+' with ')+('. So the subset $1, 3, 4 \subset [8]$ corresponds to the string

$$(1) + (1+1) + (1) + (1+1+1+1).$$

Then collapse the bracketed sums: $1+2+1+4$.

One can check that this is a bijection.

**Problem 3.7.** Using the ordering of $\mathscr{P}([7])$ you get from the bijection of it to the 7-digit binary strings, give the first three compositions of [8].

### 3.3.3 $k$ - to - 1 functions

Generalising the idea of counting by bijections or 1-to-1 functions, we can count by $k$-to-1 functions.

Given two sets $A$ and $B$, if we can define a function $f : A \to B$ such that for each $b$ in $B$, there are $k$ elements $a$ of $A$ with $f(a) = b$, then $|A| = k|B|$.

We saw this in counting the number of combinations of a set, though we didn't explicitly talk about a function.

To count the number of $k$-combinations of $n$ we define a function

$$f : (x_1, x_2, \ldots, x_k) \mapsto \{x_1, x_2, \ldots, x_k\}$$

from the set of $k$-permutations to the set of $k$-combinations. For each $k$-combination $C$, there are $k!$ different $k$-permutations $\sigma$ with $f(\sigma) = C$. So $f$ is $k!$-to-1.

We concluded that the number $C(n, k)$ of $k$-combinations was

$$C(n, k) = \frac{P(n, k)}{k!} = \frac{n!}{k! \cdots (n-k)!}.$$

### 3.3.4 The Pigeonhole Principle

A variation on the idea of counting by bijections it the idea that if we can define an onto function $f$ from $A$ to $B$, that is not a bijection, then $|A| > |B|$.

It is usually stated in the converse, with the following example:

> If there are $n + 1$ pigeons in $n$ pigeonholes, then some hole contains more than one pigeon.

This is a fairly obvious example, but the pigeonhole priciple can be used in more complicated problems.

**Problem 3.8.** Show that for any 5 points in the equilateral triangle of side length 1, there must be two whose distance apart is at most 1/2.

**Example 3.4.** A golfer plays 132 games over 11 weeks, playing at least one game a day. Show that there is some stretch of consecutive days in which she plays exactly 21 games.

**Solution**

Let $a_i$ be the number of games she plays by the end of the $i^{th}$ day. The 154 numbers $a_1, a_2, \ldots, a_{121}$, and $b_1 = a_1 + 21, b_2 = a_2 + 21, \ldots, b_{121} = a_{121} + 21$ are all between 1 and 153, so some two of them are the same. By the condition that she plays at least 1 game a day, all of the $a_i$ are distinct, and all of the $b_i$ are distinct, so there are some $i$ and $j$ such that $b_i = a_j$. So the golfer plays 21 games from the $i + 1^{th}$ to the $j^{th}$ day.

**Problem 3.9.** At a party, any two people are either acquainted, or are strangers. How many people must there be at the party so that there are either some three people that are mutually acquainted, or there are some three people that are mutual strangers.

### 3.3.5 Inclusion and Exclusion

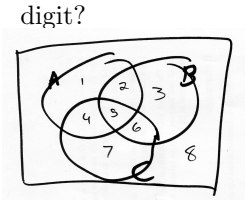
Observe from the Venn diagram that $|A \cup B| = |A| + |B| - |A \cap B|$. We can use this to count the number of integers in [60] that are multiples of 2 or 3:

**Solution**

Let $A$ be the set of multiples of 2 and $B$ be the set of multiples of 3. Then $|A| = 30$ and $|B| = 20$ and $A \cap B$ is the set of multiples of 6, so $|A \cap B| = 10$. So by the above identity we have that

$$|A \cup B| = |A| + |B| - |A \cap B| = 30 + 20 - 10 = 40$$

**Problem 3.10.** How many arrangements of the [6] start or end with an odd digit?




What about for more circles in the Venn Diagram? We see that $|A \cup B \cup C|$ is equal to

$$\begin{aligned} & |A| + |B| + |C| \\ - \quad & |A \cap B| - |A \cap C| - |B \cap C| \\ + \quad & |A \cap B \cap C| \end{aligned}$$

**Example 3.5.** Let's count the number of arrangements of the letters of the word CHEMIST that have $H$ before $T$, or $T$ before $E$, or $E$ before $M$.

Let $A_1$ be the set of arrangements with $H$ before $T$. Let $A_2$ be the set of arrangements with $T$ before $E$. Let $A_3$ be the set of arrangements with $E$ before $M$.

Then $|A_1| = \frac{7!}{2!} = |A_2| = |A_3|$, $|A_1 \cap A_2| = \frac{7!}{3!} = |A_2 \cap A_3|$, $|A_1 \cap A_3| = \frac{7!}{2!2!}$, and $|A_1 \cap A_2 \cap A_3| = \frac{7!}{4!}$.

Using the above identity, we can put this together to get $|A_1 \cap A_2 \cap A_3|$.

## 3.4 A First Word on Probability

Probability is a convenient way of counting the relative sizes of sets.

**Definition 3.6.** An *experiment* is an activity with some set of possible outcomes. The *sample space* is the set $S$ of possible outcomes. An *event* is a subset $A$ of the sample space.

**Example 3.7.** In an experiment we toss two coins. Each coin can come up 'Heads' (H) or 'Tails' (T), so our sample space is $S = \{HH, HT, TH, TT\}$. Some possibles events are

- $A_1 = \emptyset$.
- $A_2 = \{HT, TH\}$.
- $A_3 = S$.

If an experiment has a sample space in which each outcome is equally likely, the probability $Pr(A)$ of an event $A$ is

$$Pr(A) = |A|/|S|.$$

We write $Pr(a)$ for $Pr(\{a\})$.

**Example 3.8.** Continuing the previous example, we have that $Pr(A_1) = 0$, $Pr(A_2) = 1/2$, and $Pr(A_3) = 1$.

**Example 3.9.** Two distinguishable dice are rolled. Find the probability that the rolled numbers sum to a) 7, b) 9, c) 12. d) Find the probability that at least one 2 is rolled.

Solution

The sample space $S$ has 36 elements.
a) $Pr(\{16, 25, 34, 43, 52, 61\}) = \frac{6}{36} = 1/6$
b) $Pr(\{36, 45, 54, 63\}) = \frac{4}{36} = 1/9$
c) $Pr(\{66\}) = 1/36$
d) Let $A_1$ be the event that the first die is a 2 and let $A_2$ be the event that the second die is a 2. The we are looking for

$$\begin{aligned} Pr(A_1 \cup A_2) &= \frac{1}{36}|A_1 \cup A_2| \\ &= \frac{1}{36}(|A_1| + |A_2| - |A_1 \cap A_2|) \\ &= \frac{1}{36}(6 + 6 - 1) = \frac{11}{36}. \end{aligned}$$

**Problem 3.11.** You choose 2 Jellybellies from a cup of 30 different Jellybellies. What is the probability of choosing

a) banana and lemon

b) neither banana or lemon

c) banana but not lemon

**Problem 3.12.** In a group of 80 students

- 45 take Math
- 48 take CompSci
- 38 take Stats
- 30 take Math and CompSci
- 30 take CompSci and Stats
- 23 take Math and Stats
- 15 take all three

a) How many students take none of Math CompSci or Stats?

b) Find the probability that a randomly chosen student takes Math.

c) Find the probability that a randomly chosen Stats student also takes CompSci.

# 4 Induction and the Integers

In this chapter we cover all sections of Chapter 4 of the text. We also add some material about the 'modulo' function.

**Topics Covered:**

- Inductive proofs
- Recursive definitions
- The Fibonacci numbers
- Divisibility and the Division Algorithm
- Prime numbers
- The (Reverse) Euclidean Algorithm and GCD
- The Fundamental Theorem of Arithmetic
- The integers modulo $m$.

## 4.1 Induction

A *proof by induction* proceeds as follows. For a open statement $S(n)$ where the universe for $n$ is the set of positive integers, we can prove that $S(n)$ is true for all $n$ by proving the following two statements.

i. $S(1)$ is true.

ii. For all $k \geq 2$, if $S(k-1)$ is true, then $S(k)$ is true.

Having done this, the principle of induction impies that $S(n)$ is true for all $n$.

Statement 1. is called the 'basis step' and 2. is called the 'induction step'. The assumption that $S(k-1)$ is true, in the induction step, is called the 'induction hypothesis'.

See Examples 4.1 and 4.4 of the text.

In practice, the universe for $n$ may vary. But as long as there is a nice bijection between the universe of $n$ and $\mathbb{Z}^+$, we can still do induction.

See Example 4.10 of the text, for example, where the universe of $n$ is $\mathbb{N}$ instead of $\mathbb{Z}^+$. The principle of induction still applies here, indeed, we could replace every occurence of $n$ with $n-1$ and 'shift' our argument to the universe $\mathbb{Z}^+$.

### 4.1.1 Strong induction

Sometime it is useful strenghen our induction assumption, and assume that a statement $S(n)$ has been proved for more values of $n$.

See Example 4.13 of the text

## 4.2 Recursive Definitions

We can define things using a process similar to induction. This process is called recursion, it consists of a base step, and a recursion step.

The following three definitions are by recursion.

**Definition 4.1.** Let the notation $n!$ be defined as follows.

- Base Step: $0! = 1$
- Recursion Step: For $n \geq 1$, $n! = (n-1)! \cdot n$.

**Definition 4.2.** Let the general union $A_1 \cup A_2 \cup \cdots \cup A_n$ be defined for all $n \geq 2$ as follows.

- $A_1 \cup A_2 = \{x \mid x \in A_1 \vee x \in A_2\}$
- $A_1 \cup A_2 \cup \cdots \cup A_n = (A_1 \cup A_2 \cup \cdots \cup A_{n-1}) \cup A_n$

In any course in which recursion is defined, one must have the following example.

**Definition 4.3.** The $n^{th}$ Fibonacci number $F_n$ is defined for all $n \geq 0$ as follows.

- $F_0 = 1$ and $F_1 = 1$.
- For $i \geq 2$, $F_n = F_{n-2} + F_{n-1}$.

Recursive definitions lend to inductive proofs.

See Example 4.18 of the text, proving a general version of DeMorgan's Law

Recall the identity about binomial coefficients that we've proved a couple of times.

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}.$$

We can use this to recursively define the binomial coefficients as follows.

**Definition 4.4.** The binomial coefficients are defined by

- $\binom{0}{0} = 1$
- $\binom{n}{r} = 0$ if $r > n$ or $r < 0$
- $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$ otherwise